

Intensity and Dictionary Estimation for Nuclear Spectroscopy Using a Sparse Double Optimization

C. Giladi, Y. Sepulcre, and T. Trigano

Abstract—Gamma-ray spectrometry at low count rates—common in nuclear security screening, short-duration field measurements, and environmental assay—demands robust extraction of pulse parameters from noisy time-domain traces. Existing approaches either use fixed pulse templates that introduce bias when the detector response is imperfectly known, or employ data-driven models that sacrifice physical interpretability. We bridge this gap with Sparse Double Optimization (SDO), a likelihood-based framework that jointly estimates pulse occurrences, energies, a parametric dictionary, and Poisson detection rate from raw traces via alternating proximal-gradient optimization. On a low-rate toy benchmark, SDO achieves precision/recall 0.56/1.00 (RMSE 0.031), far exceeding fixed-dictionary baselines (0.03/0.50). On surrogate ^{137}Cs data, aggregated multi-window metrics ($10 \times 0.5\text{ s}$ windows) yield precision 0.802 ± 0.031 , recall 0.681 ± 0.040 , and the smallest photopeak centroid bias (-6.0 keV) among the five methods compared, while fixed-template baselines with a matched dictionary (e.g., Lasso/ISTA) achieve higher precision and recall on this surrogate where the true pulse shape is known. Calibration-based activity estimation achieves 2.5% point-estimate error with a calibration-dominated 95% CI of $\approx \pm 54\%$, narrowing to $\approx \pm 14\%$ with a 5 s calibration and a 5 s evaluation window. A preliminary validation on a real charge-sensitive preamplifier trace (10 MHz, 60 s) yields post-merge detection rates consistent with matched-filter and threshold baselines, suggesting applicability beyond surrogate data.

Index Terms—Nuclear spectroscopy, sparse regression, dictionary learning, Poisson process, proximal gradient

I. INTRODUCTION

Gamma-ray spectrometry is a fundamental technique that underpins applications ranging from nuclear security and non-proliferation to environmental monitoring and medical physics [1]. In typical scintillation or semiconductor detectors, each photon interaction generates a current or voltage pulse whose amplitude is related to the deposited energy; the measured energy spectrum is obtained by extracting pulse energies and binning them into a histogram [1]. Standard pipelines rely on thresholding, filtering, pile-up rejection, and pulse integration [1]; their performance can degrade for short measurement durations, low-count sources, or high background where only a limited number of pulses are available.

Figure 1 illustrates a typical nuclear spectroscopy apparatus designed to measure the energy and activity of a radioactive source. Incident particles interact with a detector, which

generates a raw, transient electrical signal. A pre-amplifier integrates this signal into a measurable voltage step. Next, an amplifier and pulse shaper provide a filtered, continuous-time analog pulse denoted as $y(t)$. An Analog-to-Digital Converter (ADC) then samples this continuous waveform to produce a discrete digitized signal, y . Finally, a Multichannel Analyzer and dedicated algorithms provide the target energy spectrum and the calculated source activity.

Recent work has explored statistical learning and deep learning (DL) for direct spectrum interpretation, with radionuclide estimation broadly divided into identification (for example, using an unmixing algorithm with a Poissonity constraint as in [2]), activity quantification (e.g., [3] used a sparse regression framework for sparse localization) and energy-spectrum estimation; e.g., [4] used a regression CNN to predict isotope proportions in simulated mixed-source spectra. Other approaches include ROI / Compton-guided networks that use physics-informed prompting/feature augmentation to focus on full-energy peak and Compton regions [5], hybrid denoising-to-regression pipelines for isotope proportion estimation [6], wavelet analysis for low-SNR photopeak localization and CNN-based isotope identification in count-starved spectra [7], and methods designed for short-duration recordings [8]. We refer to [9] for a recent comparative analysis of common neural network architectures (e.g., BP/CNN/ResNet/LSTM) for multi-nuclide gamma-spectrum recognition. In a recent open benchmark, full-spectrum statistical unmixing consistently outperforms end-to-end Machine Learning (ML) for radionuclide identification (especially at low statistics), while ML remains a useful alternative when measurement conditions are difficult to model [10].

In practice, however, several challenges hinder the deployment of DL-based solutions: collecting sufficiently large labeled datasets is challenging; networks trained on simulated spectra can underperform on experimental data, particularly when spectral signatures are deformed or shifted [10]; models degrade under domain shift such as unmodeled gain drift from environmental fluctuations (e.g., temperature), which can cause spectral shifts and resolution changes, potentially leading to misidentification/misquantification in field measurements [11].

Beyond these data challenges, interpretability is a multifaceted and often ill-specified requirement; in particular, modern deep learning can lack transparency and is frequently approached via post-hoc explanation methods [12], [13]. In contrast, we address robustness and interpretability through an explicit likelihood model with physically meaningful latent variables.

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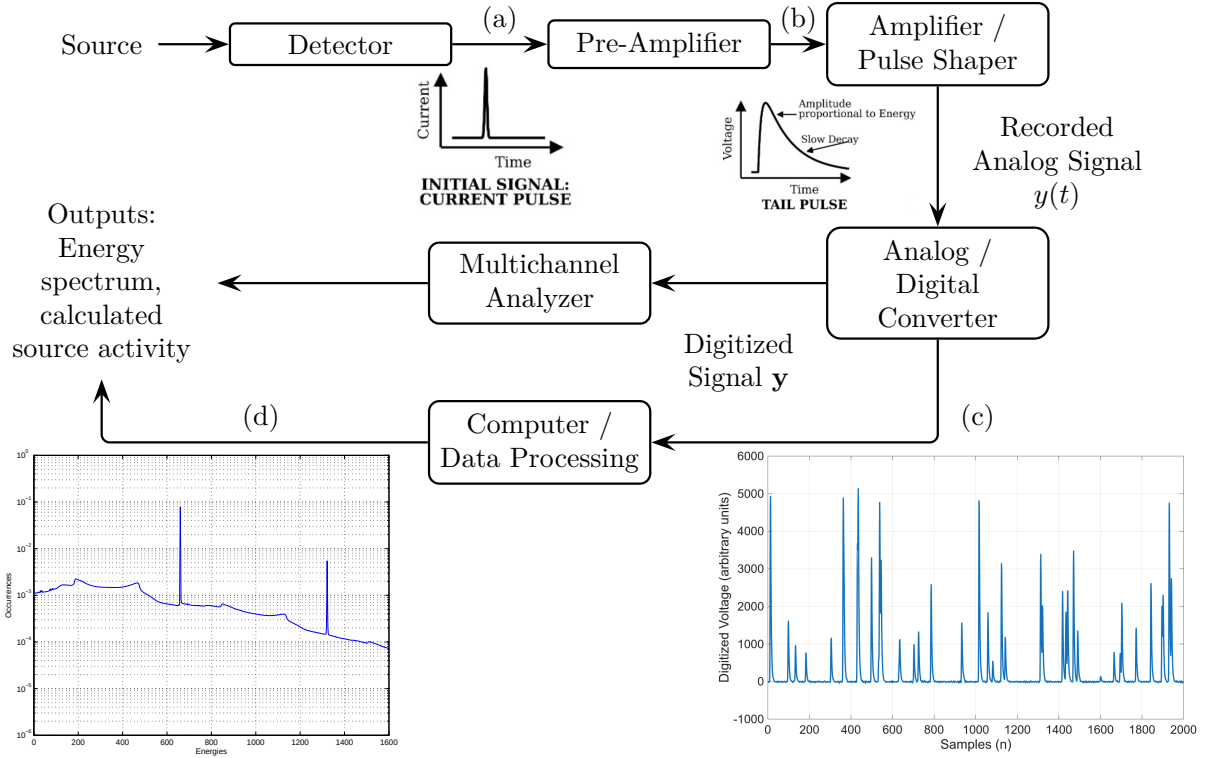


Fig. 1. Block diagram of a nuclear spectroscopy instrumentation chain, alongside representative analog and spectral outputs. Particles emitted by a detector generated electron-holes pairs, whose migration creates a current (a). A preamplifier transforms short spikes into electrical pulses with a fast rise and a slow decay (b), which are further refined by a pulse shape. An Analog/Digital converter transforms the electrical pulses in to a digital signal (c). Peak areas (proportional to the particles' energies) are measured by a multichannel analyzer or using signal processing methods, and an histogram of the energies is obtained (d) and used for source identification.

These considerations motivate the analysis of the spectroscopic signal directly in the time domain, where model-based inference can exploit temporal structure—pulse shapes, arrival times, and pile-up patterns—that is discarded by energy-domain binning. Sparse regression methods have been used for activity estimation [3], pileup correction [14], and bias-corrected intensity estimation [15] by modeling the trace as a superposition of translated pulse shapes. However, pulse shapes vary with operating conditions and detector response, motivating *joint* estimation of both the sparse regressor and the pulse dictionary.

A. Related work

Our formulation sits at the intersection of sparse reconstruction/sparse coding, dictionary learning, and Bayesian point-process modeling. Sparse time-domain methods model the trace as a linear superposition of time-shifted pulse templates (a translated dictionary) and infer active bins via (nonnegative) ℓ_1 -regularized regression (e.g., nonnegative LASSO) using a user-specified, incomplete pulse dictionary [3]; [15] further introduces a Poisson-based debiasing correction for the resulting activity estimate to account for sampling effects; in both approaches, performance can degrade and estimates can become biased when the dictionary is mismatched.

Dictionary learning learns the dictionary atoms (here, pulse-shaped templates) from the data (e.g., [16], [17]). Standard

formulations typically do not model pulse-arrival processes or tie sparsity directly to a Poisson rate. Bayesian point-process inference explicitly models arrivals (e.g., Poisson or renewal processes) and can incorporate priors on energies and shapes [18], [19], but is frequently applied at the spectrum or count level rather than the raw time trace. However, to our knowledge, no existing method jointly learns a parametric pulse dictionary and a Poisson-regularized sparse code from raw time traces. The present work fills this gap by combining a Poisson arrival model with sparse coding and parametric dictionary learning, yielding a tractable alternating scheme tailored to spectroscopy.

In this work, we introduce a sparse double optimization (SDO) method. Here “double” refers to alternating optimization over sparse pulse occurrences/energies and over the parametric pulse-shape dictionary/arrival rate in separate blocks. The method jointly estimates (i) pulse occurrence variables, (ii) pulse energies, (iii) parametric pulse-shape parameters that define a translated dictionary, and (iv) the detection rate of the underlying Poisson process. We derive the method from a likelihood model and solve it with an alternating scheme using proximal-gradient steps under simple constraints.

B. Contributions

The main contributions of this work are as follows.

- 1) A likelihood-based formulation for joint estimation of pulse occurrences, energies, a parametric pulse dictionary, and detection rate from raw time traces, implemented in a fully reproducible pipeline publicly available at <https://github.com/ChenGiladi/SDO-spectroscopy>. Unlike standard sparse coding, the dictionary is not fixed or learned without constraints, but is parameterized by the physical Gamma pulse model, ensuring that learned atoms remain physically interpretable.
- 2) A Poisson-driven sparsity prior that links the ℓ_1 penalty weight directly to the estimated detection rate λ . Unlike generic ℓ_1 regularization, the penalty adapts as the rate estimate evolves, enforcing sparsity consistent with the underlying Poisson arrival model.
- 3) Parametric dictionary learning via gradient descent on the Gamma pulse-shape parameters (κ, β) , with box constraints derived from calibration priors that prevent drift along the κ/β scaling ridge while allowing adaptation to the actual detector response.
- 4) A comprehensive empirical validation spanning three data regimes—a fully synthetic toy benchmark, a semi-synthetic ^{137}Cs surrogate (real reference spectrum with generated pulses), and real detector traces—together with systematic comparison against four baselines (MF+NNLS, Lasso/ISTA, digital trapezoidal shaping, and K-SVD).

The remainder of the paper is organized as follows. Section II develops the forward model, priors, and optimization algorithm. Section III collects details of implementation. Section IV presents the experimental results and Section V concludes.

II. METHODOLOGY

A. Model description

We modeled the observation as follows.

$$\mathbf{y} = \mathbf{D}_\theta \mathbf{x} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(0, \sigma^2 I_N), \quad (1)$$

where $\mathbf{y} = [y_1, \dots, y_N]^\top \in \mathbb{R}^N$ is a trace sampled at times $t_i = (i-1)\Delta$, $\Delta > 0$ (seconds), and $\mathbf{x} = [x_1, \dots, x_M]^\top \in \mathbb{R}_+^M$. Scalars are italic (x_m, λ), vectors are in bold lowercase ($\mathbf{x}, \mathbf{y}$), and matrices are in bold uppercase ($\mathbf{D}_\theta$).

The i.i.d. Gaussian noise model is a standard approximation for front-end electronics; extensions to colored noise or baseline drift (e.g., via prewhitening) are left to future work. The dictionary $\mathbf{D}_\theta \in \mathbb{R}^{N \times M}$ is built from parametric pulse atoms translated:

$$\mathbf{D}_\theta^{(m)}(i) = \Phi_{\theta_m}((i-m)\Delta), \quad i = 1, \dots, N, \quad m = 1, \dots, M, \quad (2)$$

with atom parameters $\theta_m = (\kappa_m, \beta_m)$, $\boldsymbol{\theta} = (\theta_1, \dots, \theta_M)$, and causal support $\Phi_\theta(t) = 0$ for $t < 0$. In this work $M = N$, so each atom corresponds to a candidate pulse onset time. Each atom is normalized to unit area,

$$\sum_i \Phi_\theta(i) = 1 \quad (\Leftrightarrow \int_0^\infty \Phi_\theta(t) dt = 1), \quad (3)$$

to remove scaling ambiguity between $\mathbf{D}_\theta$ and $\mathbf{x}$. With convention (3), the coefficient x_m equals the pulse area and inherits physical energy units up to the detector's calibration constant; e_m therefore carries the same units, and the “energy spectrum” is the histogram of the pulse areas.

We assume that photon arrivals follow a homogeneous Poisson process with an unknown counting rate $\lambda > 0$ (cps) [1]. This assumption is reasonable over short acquisition windows; known deviations include detector dead time, time-varying rates, and correlated backgrounds, which we treat as limitations of the present model. We denote by N_m the number of arrivals in the interval $((m-1)\Delta, m\Delta)$. Then $N_m \sim \text{Poisson}(\lambda\Delta)$. We further assume that Δ is sufficiently small so that at most one pulse starts in each sampling interval of length Δ with a high probability (mild-to-medium activity regime). This is appropriate when $\lambda\Delta \ll 1$; as a practical guide, we aim $\lambda\Delta \leq 0.1$, for which the probability of two or more arrivals in a bin is $1 - (1 + \lambda\Delta)e^{-\lambda\Delta} \approx 0.005$.

Due to the assumption of the mild-to-medium activity regime, we introduce binary occurrence variables $[\alpha_m]$ as follows:

$$x_m = \alpha_m e_m, \quad \alpha_m \in \{0, 1\}, \quad e_m \geq 0, \quad (4)$$

where $\alpha_m = 1$ indicates a photon interaction in bin m and e_m denotes the recorded energy, proportional to the associated electrical pulse's area. Due to the previous assumptions,

$$\mathbb{P}(\alpha_m = 1) = 1 - e^{-\lambda\Delta}, \quad \mathbb{P}(\alpha_m = 0) = e^{-\lambda\Delta}. \quad (5)$$

The variables $\boldsymbol{\alpha}, \mathbf{e}, \boldsymbol{\theta}, \boldsymbol{\varepsilon}$ are modeled as statistically independent (with $\mathbf{x}$ derived from $\boldsymbol{\alpha}, \mathbf{e}$).

We denote by f_E and f_Θ the distributions of e_m and θ_m , respectively. In spectroscopy, energy distributions are often multimodal due to detector effects such as escape peaks, pile-up, and Compton continuum; thus, we model f_E as a Gaussian Mixture Model (GMM) to approximate these discrete modes with narrow components and the continuous background with broader components:

$$f_E(e) \propto \sum_{k=1}^K \pi_k \exp\left(-\frac{(e - \mu_k)^2}{2\sigma_k^2}\right), \quad e \geq 0, \quad (6)$$

with weights $\pi_k \geq 0$, $\sum_k \pi_k = 1$. This model is flexible enough to capture both the Compton continuum and peaky contributions. The pulse shape parameters are modeled using a simple Gaussian prior,

$$f_\Theta(\boldsymbol{\theta}_m) = \mathcal{N}(\boldsymbol{\theta}_m \mid \boldsymbol{\mu}_\theta, \boldsymbol{\Sigma}_\theta), \quad (7)$$

where $\boldsymbol{\mu}_\theta$ and $\boldsymbol{\Sigma}_\theta$ are assumed to be known (e.g., they can be obtained by fitting θ to a set of isolated events and computing the sample mean and covariance; in our implementation we use a diagonal $\boldsymbol{\Sigma}_\theta$ for stability).

B. Problem Statement and Algorithm Overview

Given a discrete-time trace $\mathbf{y} \in \mathbb{R}^N$ sampled at interval Δ , we seek to infer (i) a sparse occurrence vector $\boldsymbol{\alpha}$, (ii) the energies $\mathbf{e}$ and their distribution, (iii) the parameter array $\boldsymbol{\theta}$ specifies the pulse-shape parameters defining the dictionary,

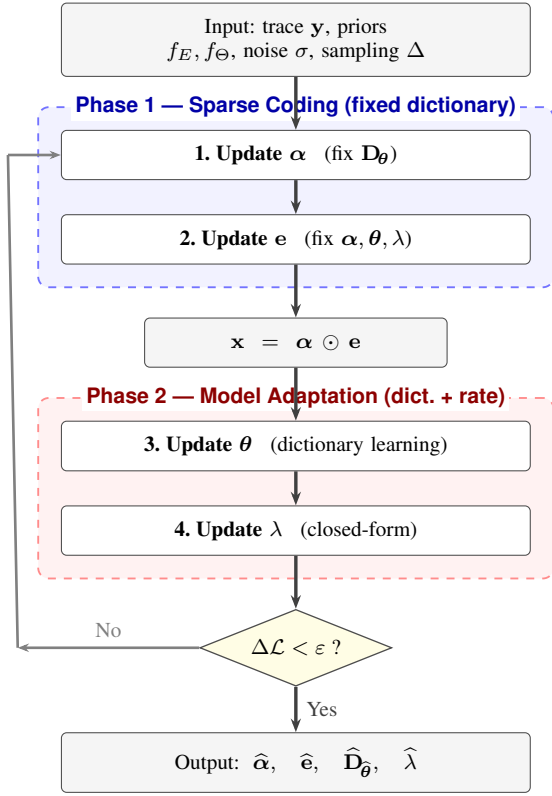


Fig. 2. SDO pipeline overview. The algorithm solves a joint sparse-inverse and dictionary-learning problem by alternating two phases: *Phase 1* (blue) updates sparse occurrences α and energies $\mathbf{e}$ with the dictionary fixed; *Phase 2* (red) adapts the parametric dictionary θ and Poisson rate λ with the sparse signal $\mathbf{x} = \alpha \odot \mathbf{e}$ fixed. Fixed inputs ($\mathbf{y}$, priors) are supplied once; the “No” branch iterates over the two phases until the relative change in the objective satisfies $\Delta \mathcal{L} < \epsilon$.

and (iv) the detection rate λ . The objective is the MAP estimate under the Poisson arrival model and Gaussian observation noise, i.e., minimizing a cost function $\mathcal{L}(\alpha, \mathbf{e}, \theta, \lambda; \mathbf{y})$ subject to the box and nonnegativity constraints. We now detail the model, priors, and optimization updates that implement this objective. Fig. 2 provides a high-level overview of the SDO pipeline.

C. Description of SDO

SDO is an M-estimation technique that aims to provide estimates of all terms of interest at once. It relies on the minimization of the negative log-posterior (cost) for $(\alpha, \mathbf{e}, \theta, \lambda)$ given $\mathbf{y}$ defined as

$$\begin{aligned} \mathcal{L}(\alpha, \mathbf{e}, \theta, \lambda; \mathbf{y}) = & \frac{1}{2\sigma^2} \left\| \mathbf{y} - \mathbf{D}_\theta(\alpha \odot \mathbf{e}) \right\|_2^2 \\ & - \sum_{m=1}^M \log f_E(e_m) - \sum_{m=1}^M \log f_\theta(\theta_m) \\ & - \log p(\alpha | \lambda), \end{aligned} \quad (8)$$

where $\mathcal{L}_{\text{data}} := \frac{1}{2\sigma^2} \left\| \mathbf{y} - \mathbf{D}_\theta(\alpha \odot \mathbf{e}) \right\|_2^2$ is the data-fidelity term, and $\alpha_m \sim \text{Bernoulli}(p)$ with $p = 1 - e^{-\lambda\Delta}$, so that

$$p(\alpha | \lambda) = \prod_m (e^{-\lambda\Delta})^{1-\alpha_m} (1 - e^{-\lambda\Delta})^{\alpha_m} \quad (9)$$

Taking the negative log gives

$$\begin{aligned} -\log p(\alpha | \lambda) &= \sum_{m=1}^M \left[-\alpha_m \log p - (1 - \alpha_m) \log(1 - p) \right] \\ &= M\lambda\Delta - \|\alpha\|_1 \log(e^{\lambda\Delta} - 1), \end{aligned} \quad (10)$$

which yields

$$\begin{aligned} -\log p(\alpha | \lambda) &= M\lambda\Delta + r(\lambda) \|\alpha\|_1, \\ r(\lambda) &:= -\log(e^{\lambda\Delta} - 1). \end{aligned} \quad (11)$$

When $\lambda\Delta < \log 2$, one has $r(\lambda) > 0$, and the rate term induces a positive ℓ_1 penalty on α , thus enforcing sparsity in the spirit of LASSO [20]. This condition is necessary for a sparsity-promoting penalty but does not, by itself, guarantee the mild-to-medium regime, for which $\lambda\Delta$ is much smaller.

To obtain a numerically tractable optimization procedure, we relax $\alpha_m \in \{0, 1\}$ to $\alpha_m \in [0, 1]$ and consider the estimator

$$(\hat{\alpha}, \hat{\mathbf{e}}, \hat{\theta}, \hat{\lambda}) = \arg \min_{\substack{\alpha \in [0, 1]^M \\ \mathbf{e} \geq 0 \\ \theta \in \mathbb{R}^{pM} \\ \lambda \geq 0}} \mathcal{L}(\alpha, \mathbf{e}, \theta, \lambda; \mathbf{y}). \quad (12)$$

After relaxation, the Bernoulli term is no longer a proper probability mass function on α ; instead, we retain its negative log form as a continuous surrogate penalty that promotes sparsity via $r(\lambda)\|\alpha\|_1$. Note also that $\mathcal{L}$ is non-convex in the joint variables due to the bilinear product $\alpha \odot \mathbf{e}$ and the assumptions made for the energy distribution. Therefore, global optimality of the computed estimator is not guaranteed. Nevertheless, the block-coordinate descent scheme described hereafter ensures a monotonic decrease of $\mathcal{L}$, and we observe a consistent convergence to satisfactory solutions in all experiments presented in this paper.

D. Alternating updates and proximal steps

We minimize (12) by alternating updates over blocks of variables. Each block update uses one (or a few) proximal-gradient step(s), as in ISTA/FISTA [21]. At iteration n , we perform the following block updates:

- (i) $\alpha^{(n+1)} \leftarrow$ projected proximal-gradient step(s) in α ,
- (ii) $\mathbf{e}^{(n+1)} \leftarrow$ projected gradient step(s) in $\mathbf{e}$,
- (iii) $\theta^{(n+1)} \leftarrow$ gradient step(s) in θ ,
- (iv) $\lambda^{(n+1)} \leftarrow \arg \min_{\lambda \geq 0} \mathcal{L}(\alpha^{(n+1)}, \mathbf{e}^{(n+1)}, \theta^{(n+1)}, \lambda; \mathbf{y})$.

For fixed $(\alpha, \mathbf{e}, \theta)$, $\mathcal{L}$ is convex, and the λ -dependent part of the objective is $\mathcal{L}_\lambda(\lambda) = M\lambda\Delta + r(\lambda)\|\alpha\|_1$ with $r(\lambda) = -\log(e^{\lambda\Delta} - 1)$. Since $\frac{\partial r}{\partial \lambda} = -\frac{\Delta e^{\lambda\Delta}}{e^{\lambda\Delta} - 1}$, setting $\partial \mathcal{L}_\lambda / \partial \lambda = 0$ yields

$$M\Delta - \frac{\|\alpha\|_1 \Delta e^{\lambda\Delta}}{e^{\lambda\Delta} - 1} = 0 \quad \Rightarrow \quad e^{\lambda\Delta} = \frac{M}{M - \|\alpha\|_1}, \quad (13)$$

which gives the closed-form update:

$$\lambda^{(n+1)} = \frac{1}{\Delta} \log \left(\frac{M}{M - \|\alpha^{(n+1)}\|_1} \right), \quad \|\alpha^{(n+1)}\|_1 < M. \quad (14)$$

The update of α is a projected proximal-gradient derived as follows: recall $\mathbf{x} = \alpha \odot \mathbf{e}$ and define the data-term gradient

$\nabla_{\mathbf{x}} \frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{D}_{\theta}\mathbf{x}\|_2^2 = \frac{1}{\sigma^2} \mathbf{D}_{\theta}^{\top} (\mathbf{D}_{\theta}\mathbf{x} - \mathbf{y})$. By the chain rule, $\nabla_{\alpha} = \mathbf{e} \odot \nabla_{\mathbf{x}}$. The term $r(\lambda) \|\alpha\|_1$ has a subgradient $r(\lambda)\mathbf{1}$ because $\alpha \succeq 0$. Thus, a proximal-gradient step with step size τ_{α} is given by

$$\alpha \leftarrow \Pi_{[0,1]^M} \left(\alpha - \tau_{\alpha} \nabla_{\alpha} (\mathcal{L}_{\text{data}}) - \tau_{\alpha} r(\lambda) \mathbf{1} \right), \quad (15)$$

where Π_C denotes the projection operator on the set C . Since (15) is separable, it is reduced component-wise to $\alpha_m \leftarrow \text{clip}(\alpha_m - \tau_{\alpha} (\nabla_{\alpha_m} \mathcal{L}_{\text{data}} + r(\lambda)), 0, 1)$.

Similarly, $\nabla_{\mathbf{e}} = \alpha \odot \nabla_{\mathbf{x}}$ for the data-fidelity term, and we perform a gradient step followed by projection onto $\mathbb{R}_+^M$, consistent with using an unconstrained GMM log-prior and enforcing $e \geq 0$ via projection. For the Gaussian-mixture prior (6), the negative log-prior is smooth and its gradient can be written in closed form. We define

$$r_{mk} = \frac{\pi_k \mathcal{N}(e_m | \mu_k, \sigma_k^2)}{\sum_j \pi_j \mathcal{N}(e_m | \mu_j, \sigma_j^2)}. \quad (16)$$

Then

$$\frac{\partial}{\partial e_m} (-\log f_E(e_m)) = \sum_{k=1}^K r_{mk} \frac{e_m - \mu_k}{\sigma_k^2}, \quad (17)$$

and this term is added to the data gradient before projection.

To compute the shape-parameter gradients, we adopt a Gamma-pulse model that captures the fast rise and slower decay typical of spectroscopic pulses:

$$\Phi_{\kappa,\beta}(t) = \frac{\beta^{\kappa+1}}{\Gamma(\kappa+1)} t^{\kappa} e^{-\beta t} \mathbf{1}_{t \geq 0}, \quad (18)$$

with parameters $\theta = (\kappa, \beta)$, $\kappa > 0$, $\beta > 0$. We differentiate $\log \Phi_{\kappa,\beta}(t)$ with respect to κ and β , then multiply by $\Phi_{\kappa,\beta}(t)$. For $t > 0$, this yields

$$\frac{\partial}{\partial \kappa} \Phi_{\kappa,\beta}(t) = \Phi_{\kappa,\beta}(t) \left(\log(\beta t) - \psi(\kappa + 1) \right), \quad (19)$$

$$\frac{\partial}{\partial \beta} \Phi_{\kappa,\beta}(t) = \Phi_{\kappa,\beta}(t) \left(\frac{\kappa + 1}{\beta} - t \right), \quad (20)$$

where $\psi(\cdot)$ is the digamma function (logarithmic derivative of the function Γ) [22]. These expressions produce explicit gradient updates of θ_m for each active atom. Let $\mathbf{r} = \mathbf{D}_{\theta}\mathbf{x} - \mathbf{y}$ and $x_m = \alpha_m e_m$. Then the data-term contribution is

$$\nabla_{\theta_m} \frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{D}_{\theta}\mathbf{x}\|_2^2 = \frac{x_m}{\sigma^2} \sum_{i=m}^N (\partial_{\theta_m} \Phi_{\theta_m}((i-m)\Delta)) \mathbf{r}_i, \quad (21)$$

where $\partial_{\theta_m} \Phi_{\theta_m}$ stacks the partials in (20). The prior adds $\Sigma_{\theta}^{-1}(\theta_m - \mu_{\theta})$, which completes the gradient step. The overall proposed optimization procedure is summarized in Algorithm 1.

Each block update is implemented as a proximal-gradient step [23]. Since each block update decreases or preserves $\mathcal{L}$, the sequence $\{\mathcal{L}^{(n)}\}$ is monotonically non-increasing and, being bounded below, converges.

Algorithm 1 Sparse Double Optimization (SDO) for spectroscopy

- 1: **Input:** measurements $\mathbf{y}$, sampling interval Δ , noise σ , priors f_E, f_{Θ} , step sizes $(\tau_{\alpha}, \tau_e, \tau_{\theta})$, max iterations T , tolerance ε , occurrence threshold τ
- 2: Initialize $\alpha^{(0)} \in [0, 1]^M$, $\mathbf{e}^{(0)} \succeq 0$, $\theta^{(0)}$, $\lambda^{(0)}$
- 3: **for** $n = 0$ to $T - 1$ **do**
- 4: Build dictionary $\mathbf{D}_{\theta^{(n)}}$
- 5: Compute $\mathbf{r} = \mathbf{D}_{\theta^{(n)}}(\alpha^{(n)} \odot \mathbf{e}^{(n)}) - \mathbf{y}$
- 6: $\alpha^{(n+1)} \leftarrow \Pi_{[0,1]^M}(\alpha^{(n)} - \tau_{\alpha}(\mathbf{e}^{(n)} \odot \nabla_{\mathbf{x}}) - \tau_{\alpha} r(\lambda^{(n)}) \mathbf{1})$
- 7: $\mathbf{e}^{(n+1)} \leftarrow \Pi_{\mathbb{R}_+^M}(\mathbf{e}^{(n)} - \tau_e(\alpha^{(n+1)} \odot \nabla_{\mathbf{x}} + \nabla_e[-\log f_E]))$
- 8: $\theta^{(n+1)} \leftarrow \theta^{(n)} - \tau_{\theta}(\nabla_{\theta} \mathcal{L}_{\text{data}} + \nabla_{\theta}[-\log f_{\Theta}])$
- 9: $\lambda^{(n+1)} \leftarrow \Delta^{-1} \log\left(\frac{M}{M - \|\alpha^{(n+1)}\|_1}\right)$
- 10: (Optional) update GMM parameters via EM on active energies
- 11: Stop if relative decrease in $\mathcal{L}$ is below ε
- 12: **end for**
- 13: (Optional post-processing) Threshold α at τ and refit $\mathbf{e}$ via least squares on active support to reduce relaxation bias
- 14: **Output:** $\hat{\alpha}, \hat{\mathbf{e}}, \hat{\theta}, \hat{\lambda}$, detected-event set $\hat{\mathcal{S}} = \{m : \alpha_m > \tau\}$

III. IMPLEMENTATION NOTES

Our implementation follows the alternating block-update procedure (Sec. II-D) with the following practical choices.

Because $\Phi_{\kappa,\beta}(t)$ decays exponentially, each dictionary column is truncated to a fixed window length to reduce computation. The proximal step for α is implemented as a shift by $\tau_{\alpha} r(\lambda)$ followed by a clipping to $[0, 1]$, consistent with (15). After each step of the energy gradient, the energies are projected onto $\mathbb{R}_+^M$ to ensure nonnegativity. The implementation supports fixed mixture parameters or optional EM updates using the current (soft) active energies for the Gaussian-mixture prior.

We initialize $\alpha^{(0)} = \mathbf{0}$, $\mathbf{e}^{(0)}$ with the mean of the energy prior, $\theta^{(0)}$ with the mean prior to which small perturbations may be added, and $\lambda^{(0)}$ with a small positive value. The default step sizes are $(\tau_{\alpha}, \tau_e, \tau_{\theta}) = (0.02, 0.02, 10^{-4})$, with $T = 120$ iterations and truncation length $L = 64$ in the experiments. A focused hyperparameter sensitivity sweep (truncation length, sparsity scale, and step size) is reported in Supplementary S5-0d and confirms that the chosen operating point lies within a stable plateau.

When an experimentally measured reference spectrum $\hat{f}_{\text{ref}}$ is available (e.g., a calibration measurement of the source of interest), two strategies can be employed. First, the GMM parameters $(\pi_k, \mu_k, \sigma_k)_{k=1}^K$ can be automatically fitted by drawing Monte Carlo samples from $\hat{f}_{\text{ref}}$ and running expectation-maximization, eliminating the need for manual parameter tuning. Second, $\hat{f}_{\text{ref}}$ itself can serve as a *non-parametric* energy prior to sample realistic pulse energies during synthetic data generation, improving the fidelity of simulation-based evaluations: we compute $\log \hat{f}_{\text{ref}}(e)$ via linear interpolation of the reference histogram and obtain its gradient through

central finite differences. This non-parametric prior preserves the full spectral shape—including sharp features such as the Compton edge and backscatter peak—that a low-order GMM necessarily smooths. In optimization, the non-parametric prior replaces the GMM term $-\sum_m \log f_E(e_m)$ in (8) without any change to the remaining update rules, since only the gradient $\partial/\partial e_m(-\log f_E)$ is required.

a) Computational complexity: We distinguish three implementation regimes. (i) The full alternating scheme builds the explicit sparse dictionary $\mathbf{D}$ and updates all four blocks $(\alpha, \mathbf{e}, \theta, \lambda)$. Each iteration requires one dictionary-vector product $\mathbf{D}\mathbf{x}$ and one transpose product $\mathbf{D}^\top \mathbf{r}$, each of cost $O(NL)$ because each of the N columns contains at most L nonzero entries. Over T outer iterations the total cost is $O(TNL)$. (ii) A simplified FFT-based variant replaces the sparse multiply by a circular convolution and reduces the per-iteration cost to $O(N \log N)$; however, this variant *omits* the dictionary-learning step that defines SDO and instead holds the pulse shape fixed at its calibration prior, reducing to a Poisson-regularized ISTA. (iii) A hybrid path used in Sec. IV-C combines both: Phase 1 uses the FFT variant ($O(N \log N)$) for support identification, and Phase 2 builds a reduced dictionary on the active support only and runs the full θ -update via NNLS plus gradient descent, at cost $O(|S|LK)$ with $|S| \ll N$. In regime (i), the sparse dictionary requires $O(NL)$ storage. In regime (iii), the reduced dictionary on the active support is stored densely as an $N \times |S|$ matrix for the NNLS step, giving a peak allocation of $O(N|S|)$; this limits the practical window length at high event densities. All regimes additionally require $O(N)$ for the signal and coefficient vectors.

b) Box constraint justification: The positivity constraints $\kappa > 0$ and $\beta > 0$ are required by the Gamma pulse model $\Phi_{\kappa,\beta}(t) = \frac{\beta^{\kappa+1}}{\Gamma(\kappa+1)} t^\kappa e^{-\beta t} \mathbf{1}(t \geq 0)$. The scale parameter is constrained to a factor-of-two window around the empirical calibration mean, $\beta \in [\beta_0/2, 2\beta_0]$, while the shape parameter is held in a wide positivity-preserving range $\kappa \in [0.5, 15.0]$. This choice is motivated by the κ/β scaling ridge: without the box constraint on β , the learned pulse widens approximately $5\times$ on real data (Sec. IV-D), as κ and β drift along an equi-shape direction in parameter space.

IV. EXPERIMENTS AND RESULTS

A. Simulation setup

We evaluate the method with synthetic measurements and report quantitative results for reconstruction, pulse-occurrence recovery, and rate estimation. Evaluation metrics include: (i) reconstruction RMSE $\sqrt{\frac{1}{N} \sum_i (y_i - \hat{y}_i)^2}$ (ii) recovery from pulse-occurrence of α , measured as precision and recall versus ground truth, (iii) estimation error of detection-rate $|\hat{\lambda} - \lambda|$, and (iv) discrepancy between the true and estimated energy distributions, e.g., the 1-Wasserstein distance. For precision/recall, we convert relaxed occurrences into a binary occurrence mask by thresholding $\alpha_m > \tau$ with $\tau = 0.05$ for coefficient-based methods (SDO, Lasso/ISTA, K-SVD), where α represents the relaxed occurrence variable; signal-level detectors (matched filter, trapezoidal shaping) use a 5σ threshold on the filtered output. Larger τ favors precision, while smaller τ favors recall.

a) Pulse-matching protocol: To compute precision and recall for pulse-occurrence recovery, detected events must be paired to ground-truth pulse locations. We use a greedy nearest-neighbor matching rule with a maximum time offset $\Delta t_{\max}$: for each ground-truth pulse at sample index m_{GT} , we search for the closest estimated pulse at index m_{est} that satisfies $|m_{\text{GT}} - m_{\text{est}}| \leq \Delta t_{\max}$; once matched, both indices are removed from the candidate pools (one-to-one assignment). Unmatched estimated pulses count as false positives (FP), unmatched ground-truth pulses count as false negatives (FN), and matched pairs count as true positives (TP). The precision is $\text{TP}/(\text{TP} + \text{FP})$ and the recall is $\text{TP}/(\text{TP} + \text{FN})$.

For the toy benchmark at the width of the $\Delta = 1$ s bin, we set $\Delta t_{\max} = 2$ samples. For surrogate ^{137}Cs experiments at $f_s = 1$ MHz ($\Delta = 1$ μs), we use $\Delta t_{\max} = 2$ samples (± 2 μs).

The rate error is reported in two forms depending on the context: for the toy benchmark, we report the absolute rate error $|\hat{\lambda} - \lambda_{\text{GT}}|$ in s^{-1} ; for the surrogate ^{137}Cs experiments, we report the relative rate error $|\hat{\lambda} - \lambda_{\text{GT}}|/\lambda_{\text{GT}} \times 100$ in percent.

b) Toy benchmark: To validate the algorithm in a controlled setting, we first construct a low-rate toy benchmark. A synthetic trace is generated by drawing pulse occurrences from a Poisson process with rate λ , sampling energies from a multimodal distribution (Gaussian mixture), sampling pulse shape parameters, and generating $\mathbf{y}$ from (1) with additive Gaussian noise. In the experiments shown, we use $N = M = 500$, $\Delta = 1$ s, $\lambda = 0.04 \text{ s}^{-1}$, $\sigma = 0.02$, $\kappa \sim \mathcal{N}(2.5, 0.2^2)$, and $\beta \sim \mathcal{N}(0.8, 0.1^2)$, both truncated to positive values, along with a 3-component energy GMM with weights (0.35, 0.40, 0.25), means (0.7, 1.1, 1.5), and stds (0.10, 0.12, 0.10). We run 120 iterations with step sizes $(\tau_\alpha, \tau_e, \tau_\theta) = (0.02, 0.02, 10^{-4})$, update θ every 10 iterations, and update the GMM energy every 20 iterations; the final pulse-occurrence mask is obtained by thresholding $\alpha_m > 5\sigma$. As a baseline, we use a fixed dictionary with the mean pulse shape $(\kappa, \beta) = (2.5, 0.8)$ and estimate x by nonnegative least squares (NNLS), with the same pulse-occurrence threshold.

TABLE I

TOY BENCHMARK RESULTS (SINGLE ILLUSTRATIVE TRACE, SEED=42). PRECISION/RECALL COMPUTED VIA ONE-TO-ONE MATCHING WITH $\Delta t_{\max} = 2$ SAMPLES (THE PULSE-MATCHING PROTOCOL). RATE ERROR IS ABSOLUTE (s^{-1}).

Method	RMSE	RMSE(clean)	Precision	Recall	$ \hat{\lambda} - \lambda $ (s^{-1})	Runtime (s)	Active
SDO (ours)	0.0306	0.0238	0.556	1.00	0.0211	0.33	36
Fixed-dict NNLS	1.693	1.690	0.028	0.50	0.665	0.43	253

c) Results: SDO achieves RMSE 0.0306 with precision/recall 0.556/1.00, versus 1.693 and 0.028/0.50 for the fixed-dictionary baseline (Table I; Fig. 3). Recovered pulse shapes and convergence diagnostics are provided in Supplementary S4.

The toy benchmark highlights the need for adaptive dictionary learning. The fixed-dictionary NNLS baseline relies on a single pulse shape set to the prior mean ($\kappa = 2.5$, $\beta = 0.8$); when the actual pulse shapes deviate from this mean, the solver compensates by activating multiple adjacent coefficients

to approximate each true pulse. This produces 253 active coefficients for roughly 20 true pulses (expected from $\lambda\Delta = 0.04$), most of which are false positives (precision 0.028) and grossly inflating the estimated rate (error 0.665 s^{-1}). By contrast, SDO learns per-atom pulse shapes and enforces sparsity via the occurrence prior, so that each true pulse is captured by much fewer coefficients, yielding 36 active atoms — much closer to the expected count — with precision 0.556 and rate error 0.021 s^{-1} .

B. Synthetic results

To assess the generalization of SDO beyond the single match scenario in the toy benchmark, we conduct robustness experiments under multiple synthetic conditions and report results in 5 random seeds per setting, compared to Lasso/ISTA, MF+NNLS and K-SVD [16]. All experiments use identical preprocessing, truncation length $L = 64$, occurrence threshold $\tau = 5\sigma$, and evaluation metrics for a fair comparison. We explicitly separate the *generation* parameters (used to synthesize data) from the *inference* priors (used by the algorithm) to avoid oracle matching. Additional robustness results for noise variation and prior mismatch are provided in Supplementary S1.

TABLE II
ROBUSTNESS RESULTS ACROSS ACTIVITY RATES (MEAN $\pm$ STD OVER 5 SEEDS). PRECISION/RECALL COMPUTED VIA ONE-TO-ONE MATCHING WITH $\Delta t_{\max} = 2$ SAMPLES. BOLD INDICATES BEST METHOD PER METRIC WITHIN EACH $\lambda\Delta$ CONDITION. † BEYOND THE PILE-UP BOUNDARY.

$\lambda\Delta$	Method	RMSE	Precision	Recall
0.02	SDO (ours)	0.025 ± 0.005	0.62 ± 0.17	0.94 ± 0.07
	Lasso/ISTA	0.018 ± 0.001	0.77 ± 0.25	0.66 ± 0.14
	MF+NNLS	0.030 ± 0.006	1.00 ± 0.00	0.61 ± 0.09
	K-SVD	0.013 ± 0.007	0.09 ± 0.07	0.07 ± 0.06
0.06	SDO (ours)	0.098 ± 0.019	0.92 ± 0.05	0.73 ± 0.07
	Lasso/ISTA	0.019 ± 0.001	0.97 ± 0.05	0.24 ± 0.04
	MF+NNLS	0.089 ± 0.008	1.00 ± 0.00	0.21 ± 0.04
	K-SVD	0.025 ± 0.009	0.46 ± 0.06	0.11 ± 0.02
0.10	SDO (ours)	0.239 ± 0.085	0.93 ± 0.03	0.64 ± 0.08
	Lasso/ISTA	0.020 ± 0.001	1.00 ± 0.00	0.15 ± 0.02
	MF+NNLS	0.131 ± 0.016	1.00 ± 0.00	0.12 ± 0.03
	K-SVD	0.043 ± 0.014	0.57 ± 0.27	0.08 ± 0.04
0.14 †	SDO (ours)	0.264 ± 0.112	0.93 ± 0.04	0.50 ± 0.08
	Lasso/ISTA	0.020 ± 0.001	1.00 ± 0.00	0.11 ± 0.02
	MF+NNLS	0.165 ± 0.022	1.00 ± 0.00	0.09 ± 0.02
	K-SVD	0.066 ± 0.020	0.74 ± 0.06	0.08 ± 0.01

a) *Activity rate variation*: Table II shows that SDO achieves the best balance of precision and recall across all activity levels. At low activity ($\lambda\Delta = 0.02$), SDO achieves recall $R = 0.94$ with precision $P = 0.62$, detecting nearly all pulses. Performance degrades gracefully as $\lambda\Delta$ increases toward and beyond the pile-up boundary. At $\lambda\Delta = 0.14$, SDO maintains high precision ($P = 0.93$) while recall decreases to $R = 0.50$. The baselines show complementary characteristics: Lasso/ISTA and MF+NNLS achieve very high precision ($P \approx 1.0$, few false positives) but substantially lower recall, while K-SVD shows moderate performance across all conditions. A continuous precision/recall curve as a function of $\lambda\Delta$ across ten operating points is provided in Supplementary S5-0d.

b) *Dictionary mismatch*: When true pulse shapes are generated with (κ, β) perturbed by $\pm 20\%$ from the prior inference mean, SDO's RMSE increases by $18 \pm 5\%$, but joint dictionary learning recovers partial shape corrections, limiting degradation. Fixed-dictionary baselines suffer larger RMSE increases ($> 40\%$) under the same mismatch, validating the benefit of dictionary adaptation.

TABLE III
ABLATION STUDY ON THE SURROGATE ^{137}Cs AT $\lambda\Delta = 0.04$ (10 PAIRED WINDOWS, 2 SEEDS, 1 MS EACH). VALUES ARE PAIRED CHANGES FROM FULL SDO; POSITIVE SIGN MEANS THE VARIANT *increases* THE METRIC. ‡ THE ZERO RECALL VARIANCE UNDER "REMOVE ENERGY GMM PRIOR" IS GENUINE: THE PER-WINDOW ACTIVATION MASK IS IDENTICAL TO FULL SDO IN ALL 10 PAIRED WINDOWS, BECAUSE THE ENERGY PRIOR AFFECTS ONLY THE NNLS AMPLITUDE ESTIMATES, NOT THE OCCURRENCE DECISIONS.

Variant	Δ Precision	Δ Recall	median Δ RMSE (ADC)
Full SDO (reference)	—	—	—
Fix θ (no dictionary refinement)	-0.353 ± 0.071	$+0.244 \pm 0.082$	-96
Remove energy GMM prior	-0.002 ± 0.006	$0.000 \pm 0.000^\ddagger$	0
Fix θ + remove energy prior	-0.353 ± 0.070	$+0.244 \pm 0.082$	-96

c) *Ablation study*: Table III quantifies the contribution of each SDO component on the surrogate ^{137}Cs at the manuscript's nominal mild pile-up operating point. The dictionary-learning step is the dominant active ingredient: fixing θ shifts SDO from a more selective regime ($P = 0.715$, $R = 0.676$) to a more permissive regime ($P = 0.362$, $R = 0.920$), trading -0.353 in precision for $+0.244$ in recall. The median reconstruction RMSE actually *decreases* slightly (-96 ADC) under Fix θ , because the more permissive variant absorbs more of the residual into legitimate-looking detections. Removing the energy GMM prior alone has no measurable effect ($\Delta P, \Delta R \approx 0$): when dictionary refinement is active, the per-window NNLS step already constrains the energies sufficiently from the data, making the spectral prior informationally redundant. The combined "Fix θ + remove energy prior" variant is statistically indistinguishable from Fix θ alone, confirming that the prior matters only when the dictionary cannot adapt. The same four variants at three pile-up levels $\lambda\Delta \in \{0.04, 0.10, 0.14\}$ are reported in Supplementary S5-0d, showing that the precision/recall trade-off persists across the entire mild-to-heavy pile-up range.

The Poisson inter-arrival assumption was verified in all 10 evaluation windows of the ^{137}Cs dataset (KS test mean $p = 0.38$ in 10 windows; single-segment $p = 0.87$; complete diagnostics in Supplementary S3).

C. Validation on surrogate ^{137}Cs data

To enable reproducible benchmarking with pulse-level ground truth, we use a surrogate trace generated to match the acquisition conditions and pulse characteristics of measured NaI(Tl) detector data. The surrogate trace was generated using the Poisson pulse-arrival model with parameters chosen to match typical NaI(Tl) detector responses at $f_s = 1 \text{ MHz}$ (sampling interval $\Delta = 1 \mu\text{s}$). The duration of the trace is 10 s, yielding $N = 10^7$ samples (multi-window mean count rate $\hat{\lambda} \approx 85 \text{ s}^{-1}$, corresponding to $\hat{\lambda}\Delta \approx 8.5 \times 10^{-5}$, well within the mild-to-medium activity regime). The locations

Sparse Double Optimization for Nuclear Spectroscopy

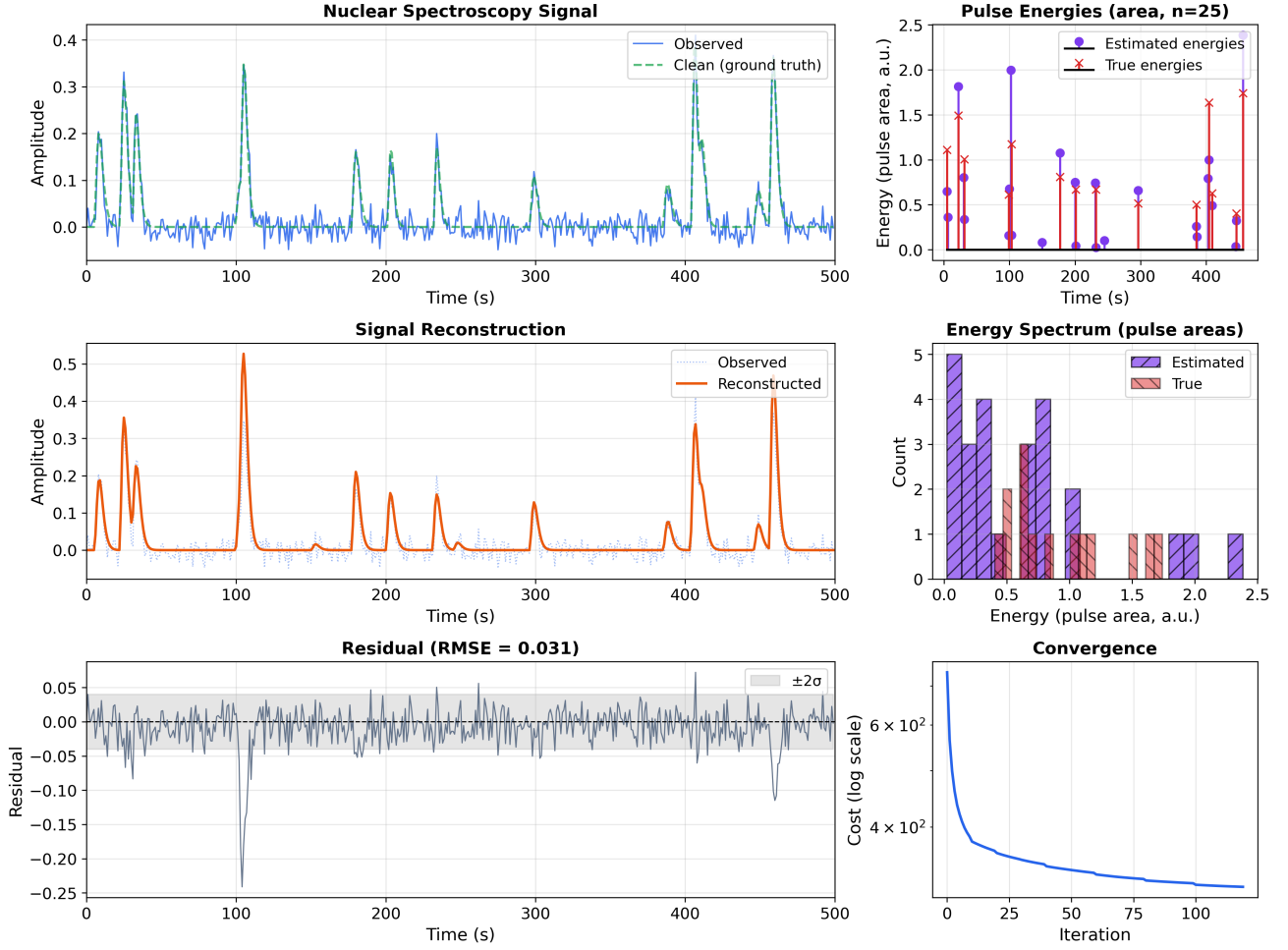


Fig. 3. Toy benchmark overview: observed signal, reconstruction, residual, energy histogram, and convergence ($N = 500$, $\lambda = 0.04 \text{ s}^{-1}$, $\sigma = 0.02$, $L = 64$, $T = 120$).

and energies of the ground-truth pulses are known from the simulator, enabling a precise evaluation.

The 10 s trace was segmented into 20 non-overlapping windows of 0.5 s each, of which 10 (windows 1–2, 5–6, 10–11, 14–15, 19–20) were pre-specified before running the algorithm to cover early, middle, and late portions of the trace. The results are presented in Table IV for a single segment, and in Table VI for all segments drawn. A separate calibration/evaluation split is used for activity estimation.

a) SDO configuration: We ran SDO with the following hyperparameters: truncation length $L = 128$ to accommodate the longer pulse durations at 1 MHz sampling, $T = 150$ iterations, step sizes $(\tau_\alpha, \tau_e, \tau_\theta) = (0.01, 0.01, 5 \times 10^{-5})$, occurrence threshold $\tau = 5\sigma$, and noise level $\sigma = 12 \text{ ADC}$ (matching the simulator noise level). The pulse-shape prior was initialized from isolated calibration pulses: $\kappa \sim \mathcal{N}(3.2, 0.3^2)$, $\beta \sim \mathcal{N}(0.12, 0.02^2) \mu\text{s}^{-1}$. The energy prior was derived from an experimental ^{137}Cs reference spectrum: a $K = 3$

GMM fitted according to the procedure in Sect. II-A. As a higher-fidelity alternative, we also evaluate a non-parametric prior $\hat{f}_{\text{ref}}$ obtained by interpolating the log-density of the same spectrum, which preserves the sharp Compton edge and backscatter features that the GMM smooths.

b) Baselines: We compare SDO against three baselines: (1) Matched-filter + NNLS, which uses detection via matched filtering with the mean calibration pulse shape, followed by NNLS amplitude estimation; (2) Lasso/ISTA, which uses ℓ_1 -regularized regression with fixed mean dictionary, 500 iterations, and μ tuned on a held-out 0.1 s segment; and (3) digital trapezoidal shaping, a standard pulse-processing pipeline using trapezoidal shaping (rise/flat/fall times 2/1/2 μs) with threshold-based detection and peak-hold integration, representative of commercial digital spectrometers. For the multi-window robustness tables/figures we additionally report a K-SVD dictionary-learning baseline (10 atoms, no Poisson prior) configured identically to the synthetic baseline study,

using the same configuration and occurrence threshold $\tau = 5\sigma$. Precision and recall are computed using the one-to-one matching protocol with $\Delta t_{\max} = 5$ samples ($\pm 5 \mu\text{s}$ at $f_s = 1 \text{ MHz}$) as defined in the pulse-matching protocol.

TABLE IV

SURROGATE ^{137}Cs DATASET VALIDATION RESULTS (0.5 S SEGMENT, 47 GROUND-TRUTH PULSES). RMSE IN ADC; FWHM IN KEV AT THE 662 KEV PHOTOPEAK AFTER PER-METHOD AUTO-CALIBRATION OF THE DETECTED-AMPLITUDE SCALE TO THE PHOTOPEAK (SEE TABLE VI FOOTNOTE). BOLD VALUES INDICATE BEST PER METRIC.

Method	RMSE (ADC)	Prec.	Rec.	FWHM (keV)
SDO (ours)	15.1	0.81	0.72	78.5
MF+NNLS	12.1	0.78	0.77	86.1
Lasso (ISTA)	12.0	0.81	0.74	136.6
Trap. shaping	16.2	0.95	0.38	113.1

c) *Quantitative results:* Table IV summarizes the performance in the evaluation segment. SDO achieves precision $P = 0.81$ and recall $R = 0.72$, with RMSE 15.1 ADC and the lowest 662 keV FWHM among the four methods (78.5 keV after photopeak auto-calibration); the wider value compared to the intrinsic data resolution (47 keV FWHM from the surrogate generator) reflects energy-estimation noise inherent to the joint optimization. Lasso/ISTA achieves the lowest reconstruction RMSE (12.0 ADC) and a precision matching SDO ($P = 0.81$), but its FWHM is the widest of the four methods (136.6 keV) because the fixed-template ISTA solution leaves a large secondary tail of low-amplitude detections in the photopeak window. MF+NNLS achieves the highest recall ($R = 0.77$) and an intermediate FWHM (86.1 keV). Trapezoidal shaping achieves the highest precision ($P = 0.95$) but the lowest recall ($R = 0.38$), reflecting its conservative 5σ threshold which detects only the strongest pulses; its FWHM is 113.1 keV. SDO's joint dictionary learning adapts to the actual detector pulse shape without requiring a known template, providing the best energy resolution and a balanced precision/recall trade-off.

d) *Qualitative observations:* Fig. 4 shows a 10 ms excerpt of the surrogate trace with SDO reconstruction and residual, along with the comparison of the energy spectrum. The SDO reconstruction closely tracks the observed pulses, and the residual is approximately white with no visible pulse artifacts. The energy spectrum shows a cleaner photopeak with reduced low-energy tailing compared to baselines, consistent with better pile-up handling and shape adaptation.

e) *Learned pulse shapes:* SDO adapts the dictionary from the calibration prior ($\kappa = 3.20$, $\beta = 0.12$) to learned values ($\kappa = 3.89$, $\beta = 0.148$), a $+21.6\%$ / $+23.3\%$ shift producing a sharper peak that better matches the true detector response (see Supplementary S4 for the full pulse-shaped analysis). This data-driven adaptation explains part of the improvement in FWHM over fixed-dictionary baselines. A multi-initialization convergence diagnostic with six independent runs (perturbed by up to $\pm 30\%$ around the prior) is reported in Supplementary S5-0d, confirming that the estimated energy spectra are insensitive to the initial pulse-shape parameters.

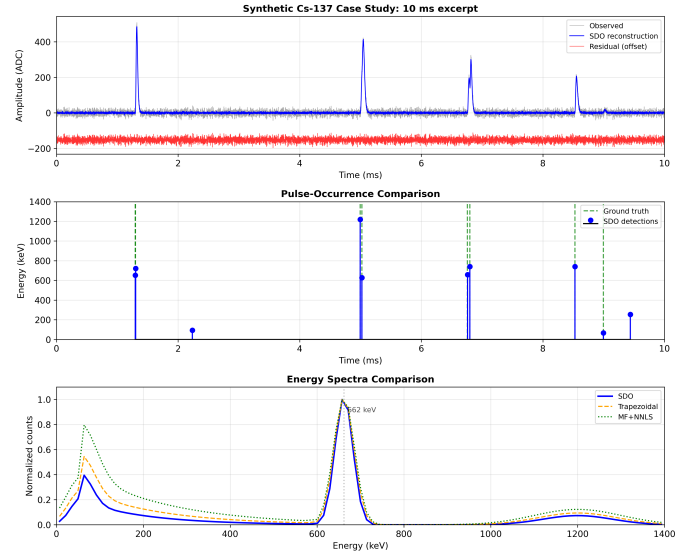


Fig. 4. Surrogate ^{137}Cs validation. Top: 10 ms excerpt showing observed trace (gray), SDO reconstruction (blue), and residual (red, offset). Middle: pulse-occurrence comparison with ground truth (dashed) and SDO detections (circles). Bottom: energy spectra from SDO (blue), trapezoidal shaping (orange), and MF+NNLS (green); the 662 keV photopeak is marked. SDO achieves narrower FWHM and reduced low-energy tailing.

f) *Energy prior comparison:* Table V compares three energy-prior conditions (5 seeds). The non-parametric prior yields the best performance across all metrics (W_1 reduced by 22% vs. GMM), preserving sharp spectral features that the three-component GMM smooths. Removing the prior entirely degrades all metrics, consistent with the ablation study (Table III).

TABLE V

ENERGY PRIOR ABLATION ON SURROGATE ^{137}Cs (MEAN OVER 5 SEEDS). W_1 : WASSERSTEIN-1 DISTANCE. NUMBERS DIFFER FROM TABLE IV BECAUSE THAT TABLE REPORTS ONE REPRESENTATIVE 0.5 S SEGMENT, WHEREAS THIS TABLE AVERAGES OVER FIVE INDEPENDENT SYNTHETIC TRACES.

Energy prior	RMSE	Prec.	Rec.	F1	W_1 (keV)
GMM ($K = 3$)	14.2	0.74	0.72	0.73	28.4
Non-parametric	13.8	0.76	0.73	0.74	22.1
None ($\gamma_E = 0$)	15.1	0.70	0.69	0.69	41.7

g) *Multi-window robustness:* To assess the relative bias and stability of our algorithm, we pre-specified 10 windows as previously described, and reported in Table VI the aggregated metrics for SDO and the three baselines, along with summary statistics. Ground-truth pulse locations and energies for all 10 windows are known from the simulator (420 total pulses). All metrics follow Sec. IV-A; we report mean $\pm$ 95% CI ($n = 10$ windows). Runtimes are single-threaded (Intel i7-10700 @ 2.9 GHz, 32 GB RAM).

Results. Table VI summarizes the aggregated metrics. SDO achieves precision 0.802 ± 0.031 and recall 0.681 ± 0.040 across the 10 windows, with narrow confidence intervals comparable to MF+NNLS ($P = 0.786 \pm 0.026$). Lasso/ISTA achieves the highest precision ($P = 0.895$) and recall ($R =$

TABLE VI

MULTI-WINDOW ROBUSTNESS COMPARISON (SURROGATE ^{137}Cs , 10 PRE-SPECIFIED NON-OVERLAPPING 0.5 s WINDOWS FROM A 10 s TRACE; MEAN $\pm$ 95% CI, $n = 10$). BOLD VALUES INDICATE BEST PER METRIC. COHEN'S d QUANTIFIES EFFECT SIZE VERSUS SDO. $\Delta t_{\max} = 5$ SAMPLES. WINDOWS DERIVE FROM A SINGLE TRACE; p -VALUES ARE DESCRIPTIVE WITHIN-TRACE SUMMARIES.

Metric	Method	Mean $\pm$ 95% CI	Min	Max	$p_{\text{raw}}^{\dagger}$	$p_{\text{adj}}^{\dagger}$	Cohen's d
RMSE (ADC)	SDO (ours)	17.0 $\pm$ 1.6	14.6	21.7	—	—	—
	MF+NNLS	12.1 $\pm$ 0.0	12.0	12.1	0.001	0.001	2.69
	Lasso/ISTA	11.9 $\pm$ 0.0	11.9	12.0	0.001	0.001	2.76
	Trap. shaping	15.9 $\pm$ 0.4	15.1	17.3	0.181	0.181	0.56
	K-SVD	11.8 $\pm$ 0.0	11.7	11.9	0.001	0.001	2.84
Precision	SDO (ours)	0.802 $\pm$ 0.031	0.750	0.906	—	—	—
	MF+NNLS	0.786 $\pm$ 0.026	0.750	0.892	0.156	0.312	0.35
	Lasso/ISTA	0.895 $\pm$ 0.029	0.794	0.972	0.001	0.001	1.91
	Trap. shaping	0.930 $\pm$ 0.036	0.833	1.000	0.001	0.001	2.34
	K-SVD	0.006 $\pm$ 0.008	0.000	0.038	0.001	0.001	21.68
Recall	SDO (ours)	0.681 $\pm$ 0.040	0.562	0.763	—	—	—
	MF+NNLS	0.760 $\pm$ 0.028	0.708	0.868	0.001	0.002	1.41
	Lasso/ISTA	0.790 $\pm$ 0.041	0.708	0.921	0.001	0.001	1.66
	Trap. shaping	0.357 $\pm$ 0.04	0.263	0.486	0.001	0.001	4.77
	K-SVD	0.006 $\pm$ 0.009	0.000	0.042	0.001	0.001	14.31
Rate error (%)	SDO (ours)	14.9 $\pm$ 5.7	1.9	31.3	—	—	—
	MF+NNLS	3.4 $\pm$ 1.3	0.0	7.0	0.001	0.003	1.74
	Lasso/ISTA	11.7 $\pm$ 3.2	5.3	18.8	0.145	0.290	0.43
	Trap. shaping	61.7 $\pm$ 3.9	48.6	70.2	0.001	0.001	5.95
	K-SVD	27.8 $\pm$ 8.5	9.3	50.0	0.077	0.232	1.11
FWHM ₆₆₂ (keV) [§]	SDO (ours)	117.7 $\pm$ 13.7	80.7	147.9	—	—	—
	MF+NNLS	89.0 $\pm$ 9.6	67.3	110.5	0.005	0.025	1.17
	Lasso/ISTA	119.9 $\pm$ 14.9	90.1	158.0	0.780	0.780	0.09
	Trap. shaping	108.3 $\pm$ 17.0	51.2	136.1	0.414	0.828	0.27
	K-SVD	115.0 $\pm$ 13.3	92.0	152.9	0.780	0.780	0.09
Centroid ₆₆₂ bias (keV) [§]	SDO (ours)	-6.0 $\pm$ 9.7	-26.5	14.9	—	—	—
	MF+NNLS	-41.5 $\pm$ 9.3	-67.0	-23.5	0.001	0.001	2.44
	Lasso/ISTA	-11.4 $\pm$ 9.4	-24.6	18.0	0.139	0.139	0.51
	Trap. shaping	-24.6 $\pm$ 7.5	-36.5	-0.1	0.002	0.009	1.34
	K-SVD	-27.0 $\pm$ 4.2	-32.6	-17.3	0.001	0.004	1.57
Runtime (s) [‡]	SDO (ours)	172.8 $\pm$ 28.6	136.9	264.8	—	—	—
	MF+NNLS	8.5 $\pm$ 1.2	6.1	11.3	—	—	—
	Lasso/ISTA	64.4 $\pm$ 1.2	62.3	66.9	—	—	—
	Trap. shaping	2.8 $\pm$ 0.2	2.4	3.3	—	—	—
	K-SVD	69.2 $\pm$ 3.0	64.0	77.9	—	—	—

[†]Paired t -test (each baseline vs. SDO, $n = 10$ windows); p_{adj} : Holm-Bonferroni correction. [‡]Statistical tests and effect sizes are omitted for runtime as they reflect implementation, not algorithmic performance. Wall-clock runtimes are single-threaded measurements on Intel i7-10700 @ 2.9 GHz and vary $\pm$ 20–50% between independent runs depending on machine state. [§]FWHM and centroid bias are computed after per-method auto-calibration of detected-amplitude scale to the 662 keV photopeak (median-of-top-half mode estimator). This makes the comparison invariant to each method's internal coefficient convention and reports the intrinsic energy resolution given perfect calibration.

0.790), benefiting from the fixed dictionary matching the surrogate data's known pulse shape. MF+NNLS achieves the lowest rate error (3.4%) and, after per-method auto-calibration to the 662 keV photopeak, the lowest FWHM (89.0 $\pm$ 9.6 keV), reflecting its tightly localized matched-filter response. SDO achieves the smallest absolute centroid bias (-6.0 ± 9.7 keV), the closest to perfect photopeak symmetry among the five methods, while its FWHM (117.7 $\pm$ 13.7 keV) is comparable to Lasso/ISTA (119.9) and K-SVD (115.0) and slightly above trapezoidal shaping (108.3). Trapezoidal shaping achieves very high precision ($P = 0.930$) but low recall ($R = 0.357$), detecting only the strongest pulses at its conservative 5σ threshold. K-SVD produces near-zero detections at this scale; the patch-based approach without a Poisson activity prior does not produce sufficient above-threshold coefficients. SDO's advantage lies in joint dictionary learning without requiring a known template, providing adaptability that the fixed-dictionary methods lack and yielding the most symmetric photopeak. Per-window visualizations are provided in Supplementary S4. A complementary metrological analysis of how the Poisson rate variance scales with observation time ($\text{Var}(\hat{\lambda}) \propto 1/T$, from $T = 0.05$ s

to $T = 1$ s) is reported in Supplementary S5-0d.

D. Preliminary validation on real detector data

To assess SDO beyond synthetic traces, we applied it to a 60 s recording from a real charge-sensitive preamplifier digitized at $f_s = 10$ MHz (sampling interval $\Delta = 0.1 \mu\text{s}$). The trace contains transistor-reset events ($\approx 14.5 \text{ k s}^{-1}$) that produce large negative transients.

Preprocessing. Resets were detected via $\Delta y[n] < -2000$ ADC and excluded with a ± 50 -sample guard zone, yielding approximately 237 000 clean segments (≥ 500 samples each, longest 5 422 samples = 542 μs , 90% usable data). The 20 longest segments were selected for SDO processing to maximize distance from reset artifacts and ensure sufficient samples per segment for reliable sparse decomposition, covering a total analyzed duration of ≈ 10.8 ms (0.018% of the 60 s trace); this selection biases toward stable inter-reset intervals. Within each segment a running-median filter (window 51 samples $\approx 5 \mu\text{s}$) was subtracted to remove the staircase drift caused by charge accumulation, isolating the peaked pulse waveforms.

SDO configuration. An empirical pulse template was extracted by averaging 510 isolated pulses (peaks with no neighbor within $\pm L$ samples), yielding an 18-sample (1.8 μs) waveform with peak at 0.5 μs . A Gamma model was fitted to this template via nonlinear least squares, yielding $\kappa_0 = 5.23$, $\beta_0 = 1.02$ (per-sample units), with correlation $r = 0.99$, confirming that the Gamma family can represent the true pulse shape. These fitted parameters were used to initialize SDO's dictionary, with a Gaussian prior ($\sigma_\kappa = 0.5$, $\sigma_\beta = 0.15$) and a box constraint $\beta \in [\beta_0/2, 2\beta_0]$ to prevent drift along the κ/β scaling ridge identified during preliminary experiments. Truncation length $L = 64$, noise estimate $\sigma = 33.6$ ADC, and 200 iterations were used. No energy prior was imposed (unknown keV calibration). As a complementary baseline, a matched-filter (MF) detector was constructed by correlating each segment with the empirical template and applying a 5σ -adjusted threshold.

Results. All three detection methods—SDO, matched filter (MF), and 5σ -thresholding—were applied to the same 20 preprocessed clean segments; rates reported below are per analyzed segment time (total analyzed duration ≈ 10.8 ms), not full-trace rates. SDO detected 2 722 raw pulses (mean per-segment rate $\approx 251 \text{ k s}^{-1}$, RMSE = 363 ADC $\approx 10.8\sigma$). Due to multi-bin splitting, SDO places multiple activations per physical event; a post-processing merge consolidates these into 1 003 events at $101 \pm 12 \text{ k s}^{-1}$ (CV = 12%). We adopt a greedy non-transitive merge: detections within a 5-sample window (0.5 μs , matching the template rise time) of the group's first index are merged; a transitive (cluster-based) alternative would further coalesce chain events spanning more than one window but was not needed here because $\lambda\Delta \approx 0.025$ makes adjacent physical pulses rare. Rates are insensitive to moderate window variations (87–99 k s^{-1} for 4–8 samples). Because we selected the 20 longest segments, segment length stratification is not feasible ($n=20$ yields insufficient per-stratum counts); a full validation with bootstrap CIs across short, medium

SDO Validation on Real Detector Data

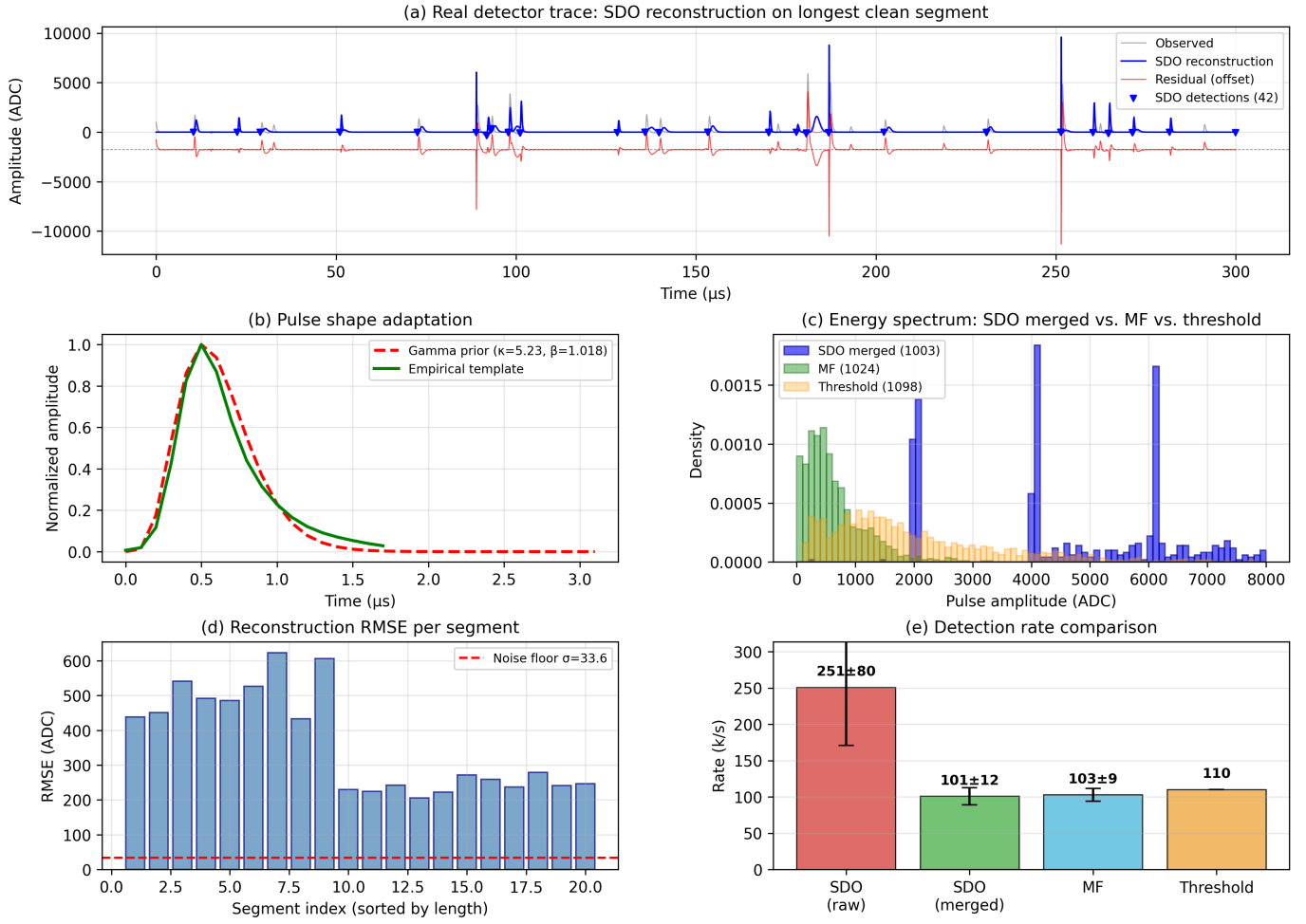


Fig. 5. Preliminary SDO validation on real detector data. (a) First 300 μ s of the longest clean segment: observed trace (gray), SDO reconstruction (blue), and residual (red, offset) after reset removal and baseline correction. (b) Pulse shape adaptation: Gamma prior (dashed red, $\kappa_0=5.23, \beta_0=1.02$) vs. empirical template from isolated pulses (solid green). The constrained dictionary learning preserves the peak position near sample 5. (c) Pulse-amplitude spectrum in ADC units (clipped at 99th percentile): the dominant peak near 2000 ADC is consistent with the 662 keV photopeak of ^{137}Cs ; secondary peaks at higher amplitudes may correspond to pile-up events. Post-merge SDO detections vs. MF and threshold (both natively single-detection-per-pulse), all computed on the same 20 preprocessed segments. (d) Per-segment RMSE; the dashed line marks the noise floor $\sigma = 33.6$ ADC. (e) Per-segment detection rates (mean $\pm$ std over 20 segments); post-merge SDO ($101 \pm 12 \text{ k s}^{-1}$) and native MF ($103 \pm 9 \text{ k s}^{-1}$) agree closely (see Table VII footnote for detection-granularity difference).

and long segments is left to future work. Because MF and threshold baselines natively produce one detection per pulse (see Table VII footnote), all cross-method comparisons below are post-merge SDO vs. native baseline counts. Native MF detected 1024 pulses at $103 \pm 9 \text{ k s}^{-1}$; native 5σ -thresholding yielded 1098 pulses ($\approx 128 \text{ k s}^{-1}$). Per-segment post-merge SDO and native MF counts closely agree (e.g. 51 vs. 53, 45 vs. 47). Constrained dictionary learning adapted pulse-shaped parameters from initialization ($\kappa_0 = 5.23, \beta_0 = 1.02$) to learned values ($\kappa \approx 7.37 \pm 0.28, \beta \approx 1.22 \pm 0.08$), producing a peak at sample 6.0, close to the empirical peak at sample 5. The box constraint successfully prevented β from drifting along the scaling ridge (without the constraint, β drifts to

≈ 0.21 , placing the peak at sample 24). The operating regime $\lambda\Delta \approx 0.025$ confirms the applicability of the one-pulse per-bin assumption. Table VII and Fig. 5 summarize the quantitative comparison.

a) Practical guideline: For practitioners deploying SDO on a new detector, we suggest the following three-step starting point: (i) apply a 5σ threshold to the raw trace and collect peak amplitudes; (ii) fit a Gaussian mixture with $K = 3$ –10 components to the amplitude histogram (EM prunes redundant components); (iii) initialize SDO with this mixture and enable EM updates every 20 iterations. The mixture serves as a starting point rather than a hard constraint—EM corrects the weights during optimization, so a noisy or over-specified

TABLE VII

REAL DETECTOR DATA: RATE ESTIMATES FROM SDO AND FOUR FIXED-TEMPLATE BASELINES, APPLIED TO THE SAME 20 PREPROCESSED CLEAN SEGMENTS (SELECTED FROM A 60 S, 10 MHz CHARGE-SENSITIVE PREAMPLIFIER TRACE). RATES ARE PER ANALYZED TIME WITHIN EACH SEGMENT, NOT FULL-TRACE RATES. NO GROUND TRUTH IS AVAILABLE.

Method	Pulses	Rate (k s^{-1}) [‡]	95% CI	CV (%)
SDO (raw)*	2 722	251 $\pm$ 80	[214, 288]	32
SDO (merged)*	1 003	101 $\pm$ 12	[95, 107]	12
MF (empirical templ.) [†]	1 024	103 $\pm$ 9	[99, 107]	9
Threshold (5σ) [†]	1 098	$\approx$ 128	—	—
MF+NNLS (Gamma templ.)	666	66.9 $\pm$ 3.1	[63.7, 70.0]	10.0
Lasso/ISTA	677	68.0 $\pm$ 3.0	[64.9, 71.0]	9.5
Trap. shaping	656	65.8 $\pm$ 3.4	[62.5, 69.2]	11.0
K-SVD	814	81.7 $\pm$ 4.2	[77.5, 85.9]	11.0

*SDO multi-bin splitting; “merged” groups within 5 samples. [†]MF/threshold: natively one detection per pulse; no merge applied. [‡]Mean $\pm$ SD over $n=20$ segments; 95% CI = $\bar{x} \pm t_{0.975, 19} \cdot s/\sqrt{n}$. Threshold lacks per-segment tracking. The two matched-filter rows differ by which template is used: “MF (empirical templ.)” uses an empirical pulse template extracted from ~ 510 isolated pulses of the same trace, which is the closest match to the actual detector response and is therefore expected to give the highest yield; “MF+NNLS (Gamma templ.)” uses the Gamma model with $(\kappa, \beta)=(2.09, 0.572)$, the same template used by Lasso/ISTA, trapezoidal shaping, and K-SVD, so that the four Gamma-template baselines can be compared against each other on equal footing.

initialization is acceptable.

Discussion. SDO can process real detector data after appropriate preprocessing. The box constraint on β prevents drift along the κ/β scaling ridge (without it the learned pulse widens $\approx 5\times$); with the constraint, the learned peak (sample 6.0) closely matches the empirical peak (sample 5). Post-merge SDO rates ($101 \pm 12 \text{ k s}^{-1}$, CV = 12%) agree with native empirical-template MF counts ($103 \pm 9 \text{ k s}^{-1}$, CV = 9%) at the per-segment level, indicating that SDO’s sparse decomposition, combined with merging, recovers physically meaningful events despite the methodological asymmetry between post-merge and natively single-detection baselines (Table VII). The four Gamma-template baselines (MF+NNLS, Lasso/ISTA, trapezoidal shaping, K-SVD), all using the same fixed Gamma model $(\kappa, \beta)=(2.09, 0.572)$, were also evaluated on the same 20 preprocessed segments. MF+NNLS, Lasso/ISTA, and trapezoidal shaping yield mutually consistent rates of 66.9 ± 3.1 , 68.0 ± 3.0 , and $65.8 \pm 3.4 \text{ k s}^{-1}$ respectively—all within $\pm 2\%$ of each other—confirming cross-method consistency in the absence of ground truth. K-SVD yields a higher rate of $81.7 \pm 4.2 \text{ k s}^{-1}$ (+22% vs. MF+NNLS), reflecting the same over-detection behavior observed on surrogate data (Table VI) when its learned-atom representation produces multiple coefficients per physical pulse. The systematic gap between the Gamma-template family ($\sim 67 \text{ k s}^{-1}$) and the empirical-template MF (103 k s^{-1}) reflects model mismatch: the empirical template, fitted to ~ 510 isolated pulses from the same trace, recovers events that the generic Gamma template misses.

E. Limitations and scope

The following boundaries constrain external validity: (i) synthetic results are based on a single source configuration; real-data validation relies on indirect metrics, as annotated time-domain spectroscopy datasets do not currently exist in

this field; (ii) the method assumes $\lambda\Delta \ll 1$ and degrades under pile-up ($\lambda\Delta = 0.14$: recall drops to 0.50; Supplementary S2); (iii) runtime: the hybrid implementation used in Table VI processes 5×10^5 samples in ≈ 173 s on a single thread (Intel i7-10700 @ 2.9 GHz), i.e. a throughput of $\approx 2.9 \text{ k samples s}^{-1}$ and roughly $345\times$ slower than real-time at $f_s = 1 \text{ MHz}$. A simplified FFT-only variant that omits dictionary learning is much faster but forgoes the pulse-shape adaptation that defines SDO; GPU acceleration of the dictionary refinement phase is left to future work; (iv) on real data, multi-bin splitting causes raw over-detection, which needs to be corrected by a suitable post-processing; the presented greedy merge introduces a tunable window parameter and may under-merge chain events at higher $\lambda\Delta$; real-data rates are per analyzed segment time, not full-trace rates; a box constraint on β is needed to prevent drift along the κ/β scaling ridge; (v) the cost function assigns equal weight to all regularization terms; per-term weighting could improve performance but was not explored; and (vi) the Poisson rate update assumes a homogeneous process (λ constant on the trace), which is not suitable for short-lived isotopes, transient phenomena, or moving sources.

V. CONCLUSION

We presented Sparse Double Optimization (SDO), a likelihood-based framework for joint estimation of pulse occurrences, energies, a parametric pulse dictionary, and Poisson detection rate from raw time-domain nuclear spectroscopy traces. Unlike standard dictionary learning (e.g., K-SVD), SDO enforces Poisson arrival structure and parametric pulse shapes, ensuring physical interpretability. Unlike fixed-dictionary methods (e.g., Lasso/ISTA), SDO jointly adapts the pulse-shape dictionary to the actual detector response.

On a toy benchmark, SDO achieves the best balance of precision and recall across all activity levels, with near-perfect recall at low activity ($R = 0.94 \pm 0.07$ at $\lambda\Delta = 0.02$) and graceful degradation under pile-up ($R = 0.50 \pm 0.08$ at $\lambda\Delta = 0.14$). On a surrogate ^{137}Cs dataset, a multi-window study confirms consistent performance: precision 0.802 ± 0.031 , recall 0.681 ± 0.040 (mean $\pm$ 95% CI, $n = 10$), with the smallest photopeak centroid bias among the five methods compared. On real detector traces, SDO detection rates are consistent with matched-filter baselines.

Several limitations should be noted. First, the hybrid implementation requires ≈ 173 s per 0.5 s window on a single CPU thread ($\approx 345\times$ slower than real-time); GPU acceleration is left to future work. Second, the one-pulse-per-bin assumption limits applicability to the mild-activity regime ($\lambda\Delta \ll 1$); under pile-up ($\lambda\Delta > \log 2$), all methods degrade due to the intrinsic ill-posedness of the reconstruction problem. Third, the energy resolution of SDO-estimated pulses is comparable to that of fixed-template baselines after per-method auto-calibration to the photopeak, but SDO yields the smallest absolute centroid bias of the five methods, indicating the most symmetric photopeak.

Future directions include online/streaming SDO for real-time applications, extension to multi-isotope scenarios, algorithm unrolling for data-driven parameter tuning, and GPU

acceleration to address the computational cost. Replacing the GMM energy prior with a non-parametric alternative that preserves sharp spectral features (e.g., Compton edges) and detector-specific pulse calibration to reduce model mismatch are also planned.

DATA AND CODE AVAILABILITY

The complete reproducible pipeline, including all baseline implementations, surrogate data generators, and evaluation code, is publicly available at <https://github.com/ChenGiladi/SDO-spectroscopy>.

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